ONE-PAGE QUICK REFERENCE · WORKING THEORY IS NOT A CONFIRMED DIAGNOSIS

STOP DRIVING IF THE MANIFOLD IS GLOWING OR OIL IS ON THE EXHAUST. COOL COMPLETELY.

Glowing manifold

STOP Bright orange or red iron. Crack, burned valves, melted cats, fire if oil is present. Do not spray water. Do not drive it. Tow if it has to move. If it glows again, stop immediately.

Causes still open: lean burn, misfire afterburn, retarded timing, restricted cat. **POSSIBLE** each. **TEST REQUIRED**

Oil / fire

STOP Oil on hot iron or on an O2 sensor can ignite. Separate problem from the idle code. Degrease only when cold. Trace fresh oil.

Common 4.0L sources: valve cover, filter-adapter O-rings, pressure sender, pan, rear main. **LIKELY** places. Repair the leak before replacing an oil-soaked sensor.

IAC — do not touch the pintle

Stepper motor. PCM counts steps and cannot see the tip. This pintle was moved. **STATED** Treat it as damaged. New motor only. Never push, pull, or pry the new pintle.

Candidate: Mopar 4874373AB. **CATALOG LISTED** Verify by VIN. Coils that ohm fine do not save a stripped screw.

Vacuum leak

MAP engine, no MAF. A leak raises idle. **LIKELY** It does not automatically make the mixture lean, because the PCM sees the MAP change and adds fuel. Lean is **POSSIBLE** and must be measured. Smoke test. No carb-cleaner spray while oil or a glow is in play. Fix leaks before the relearn.

P0505

CONFIRMED title: idle control system. Often an IAC circuit short on this family. **LIKELY** A high idle is a separate fact and does not prove the code, or a vacuum leak.

Path: RPM, leaks, IAC ohms, pintle history, new IAC, clean bore, relearn, verify. Guide 03 and Guide 08.

P0161

CONFIRMED Bank 2 Sensor 2 heater circuit. Rear downstream, after the rear mini-cat. **LIKELY** Not a mixture code. Not the cause of a high idle.

Connector, fuse, cold ohms, 10–12 V while ASD is on, PCM control. NTK 23142 and 56029001AB are **NOT CONFIRMED** for this truck.

WARM IDLE

700–800 rpm screen, Park, loads off. TSB about 744 rpm. **TEST REQUIRED**

IAC COILS

About 40–80 Ω each. Often 48–60. Isolated from each other. **LIKELY**

O2 HEATER

About 4–10 Ω cold. Earlier spec about 4–7 Ω. OL = open. **TEST REQUIRED**

HEATER POWER

About 10–12 V while ASD is on. Key-on 0 V can be a false fail.

Repair order

1. Stop if glowing or oil is on hot exhaust.
2. Cool completely. No water on the iron.
3. Find and fix the vacuum leak. Smoke, not spray.
4. New IAC, correct unit, verified by VIN.
5. Do not move the new pintle.
6. Clean the throttle body. Plate must close.
7. PCM reset and idle relearn. Foot off the pedal.
8. Verify warm idle. Record RPM.
9. Find and repair the oil leak. Fresh oil decides.
10. Retest P0161. Repair wire or sensor as the test says.
11. Clear codes only after the tests are written down.
12. Start, watch the manifold and the oil, rescan.
13. If it glows again, stop immediately and tow.

Guides 01–09 have the procedure for each line.

Possible chain, not confirmed: pintle moved → IAC damaged or held open → extra air → high idle → P0505 → mixture may go lean → exhaust may glow. Prove each step. A misfire can glow the manifold without that chain.

Two other problems stay separate: oil leak → fire on hot iron. And P0161 → heater circuit of Bank 2 Sensor 2 → test sensor, power, control, fuse, wiring. Do not trade one repair for another.